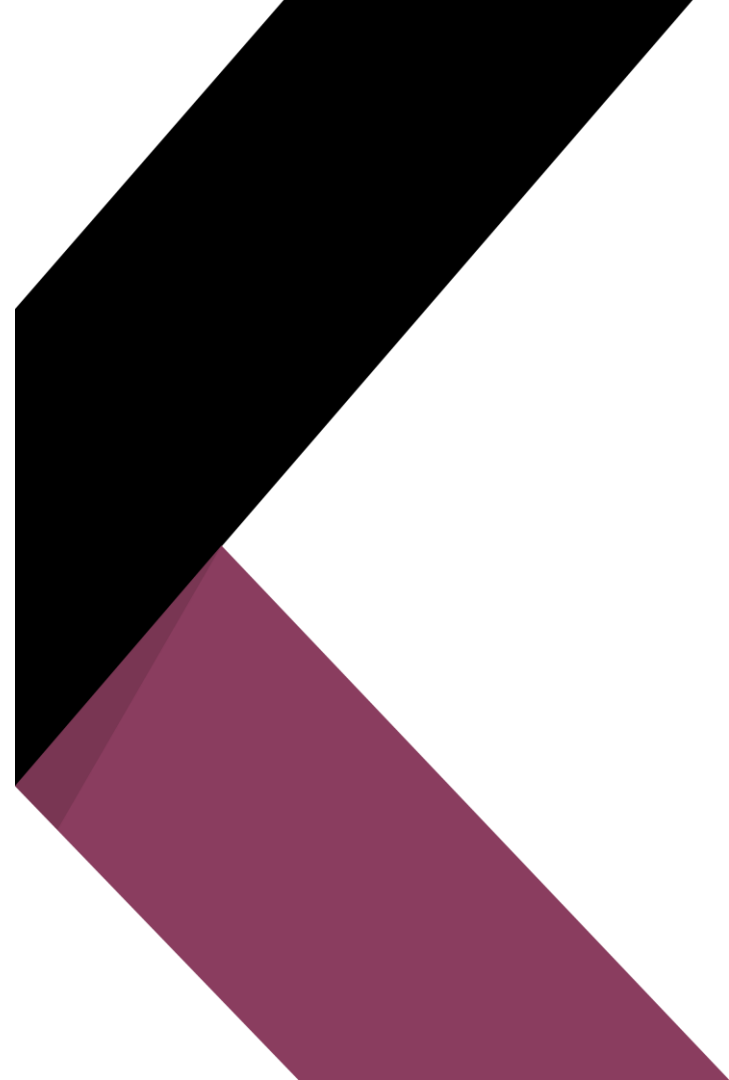


FLA

First Level Analysis



1. First Level Analysis

- ☐ FLA mission and scope
- ☐ Workflow
- ☐ Tools and Server
- ☐ Preliminary know-how

2. Case Study FLA - FTT#344 (MHEV vehicle)

- ☐ Claim visualization through signals
- ☐ Vehicle boundary conditions
- ☐ Possible root cause to address 2nd level

3. PDX use in validation and troubleshooting

- ☐ Diagnostic response decoding in CANalyzer



First Level Analysis

Team Mission & Scope:

-First Level Analysis

- ☐ To produce a report of the failures identifying all the data necessary for second level analysis in terms of DataLogger Data, Vehicle Scan Report, Vehicle Status as reported by Fleet Support
- ☐ To address the function in Propulsion organization to start second level analysis assigning a clear ownership or to EE Vehicle Team to start second level analysis in case of issue out of propulsion area
- ☐ To trace the issue evolution by means of a dedicated Open Item List (Fault Tracking Table) monitoring also new occurrences of it

FLA

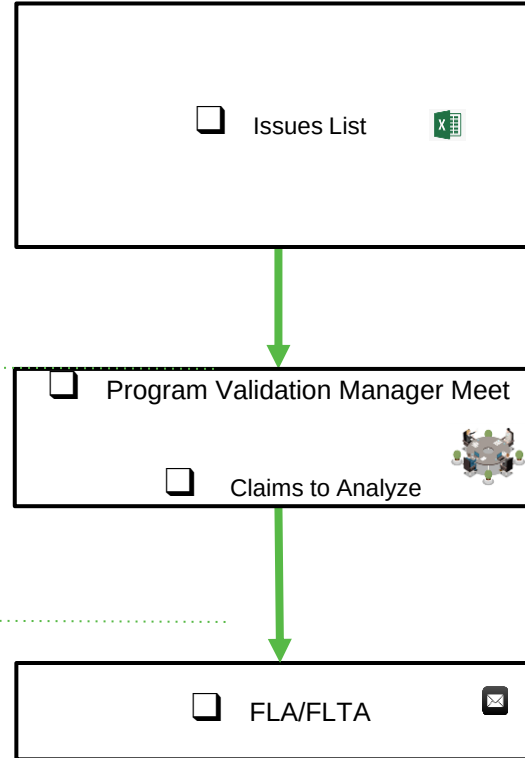
Team
Mission & Scope

Workflow

- ☐ Total Claims Received:
 - ☐ Flying Doctor and Fleet Support Engineer
 - ☐ HMSR Fleet Report
 - ☐ RG / Captive / FFB Fleet Report - IMAN or Ticket
 - ☐ End of Line claim
 - ☐ OBD Scorecard Control Tec
 - ☐ Extra (ex: SXV fleet, durability, /H, Reapper Test...)

- ☐ Sharing of issues received with PVM
- ☐ Defining of order and priority of Claims to Analyze

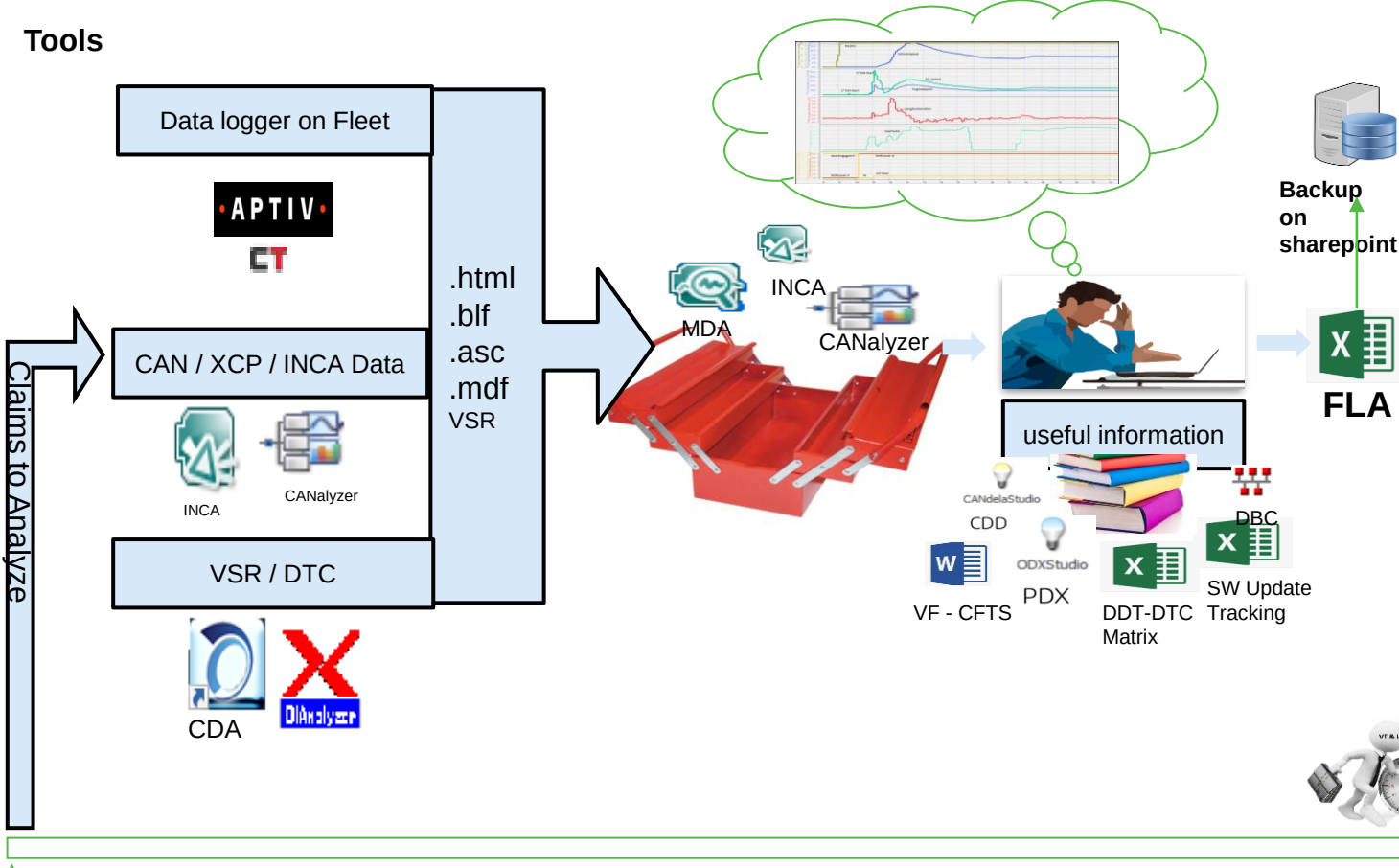
- ☐ Sharing of analysis and FLA by e-mail and data upload on sharepoint



FLA

Workflow

Tools



FLA

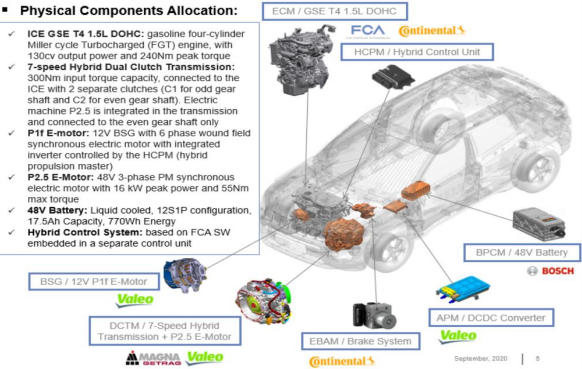
Tools

Claims to Analyze

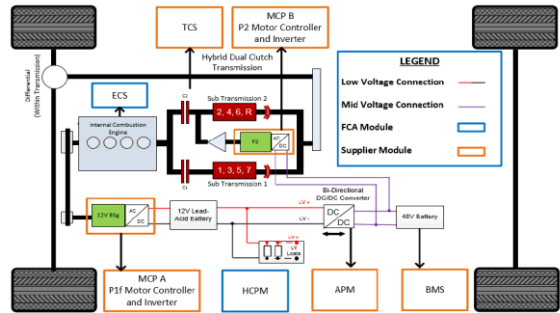
Start of FLA Process

within 24h or faster

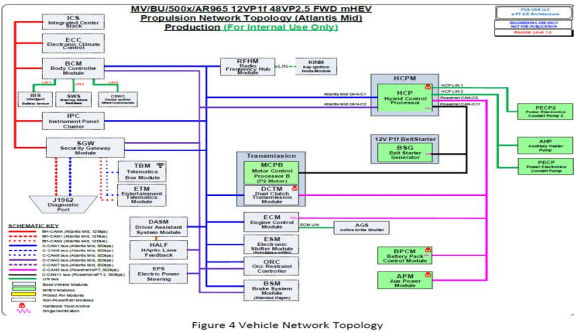
Propulsion System Configuration



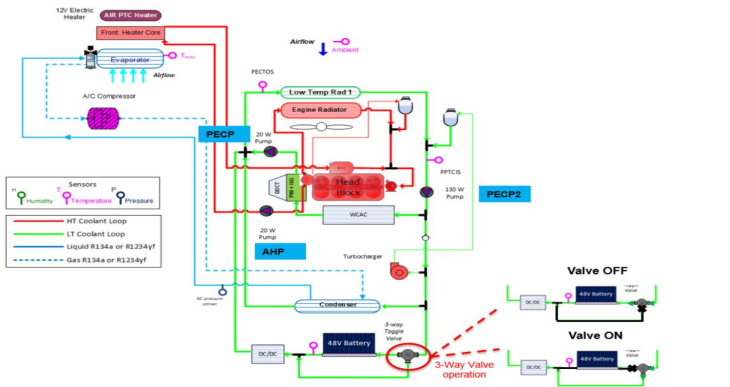
Mechanical/Electrical Layout



Network Topology



Cooling System Configuration

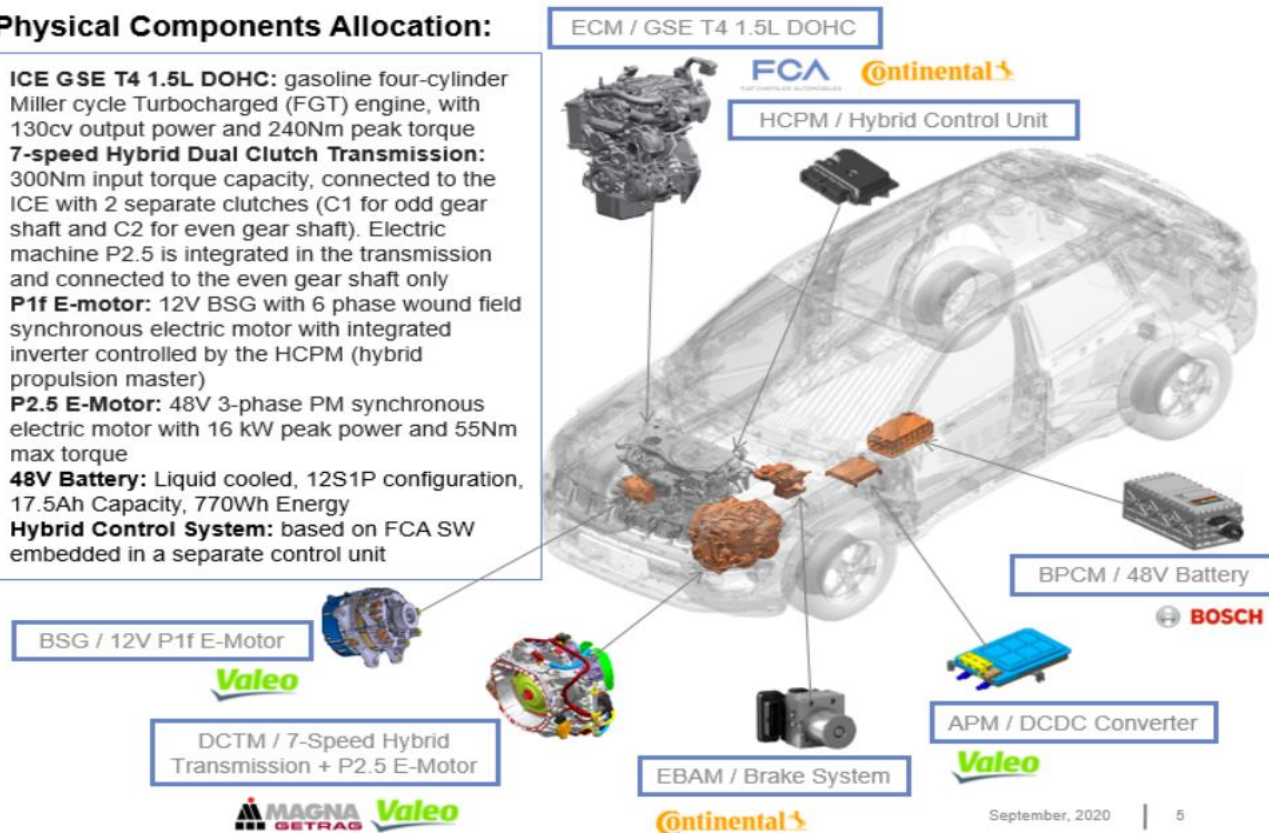


FLA

Preliminary know-how

Physical Components Allocation:

- ✓ **ICE GSE T4 1.5L DOHC:** gasoline four-cylinder Miller cycle Turbocharged (FGT) engine, with 130cv output power and 240Nm peak torque
- ✓ **7-speed Hybrid Dual Clutch Transmission:** 300Nm input torque capacity, connected to the ICE with 2 separate clutches (C1 for odd gear shaft and C2 for even gear shaft). Electric machine P2.5 is integrated in the transmission and connected to the even gear shaft only
- ✓ **P1f E-motor:** 12V BSG with 6 phase wound field synchronous electric motor with integrated inverter controlled by the HCPM (hybrid propulsion master)
- ✓ **P2.5 E-Motor:** 48V 3-phase PM synchronous electric motor with 16 kW peak power and 55Nm max torque
- ✓ **48V Battery:** Liquid cooled, 12S1P configuration, 17.5Ah Capacity, 770Wh Energy
- ✓ **Hybrid Control System:** based on FCA SW embedded in a separate control unit



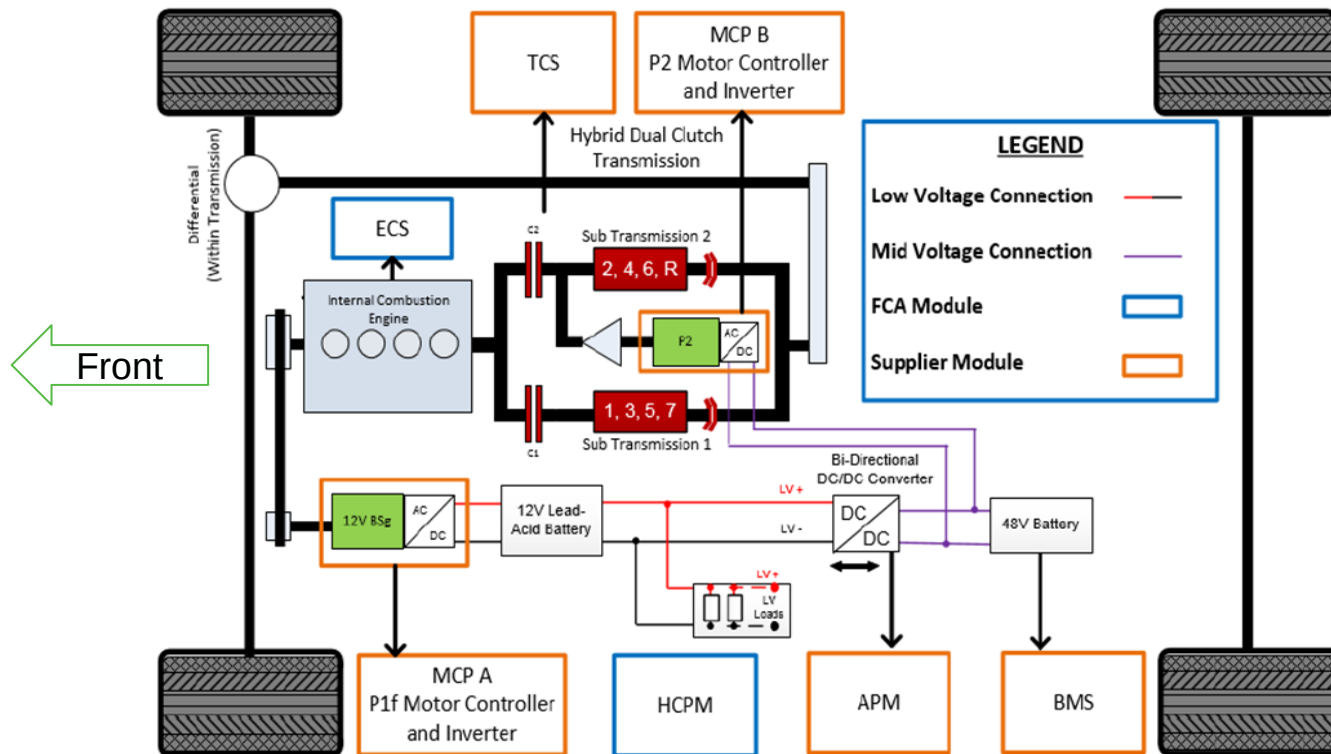


Figure 2 Propulsion System Mechanical Layout

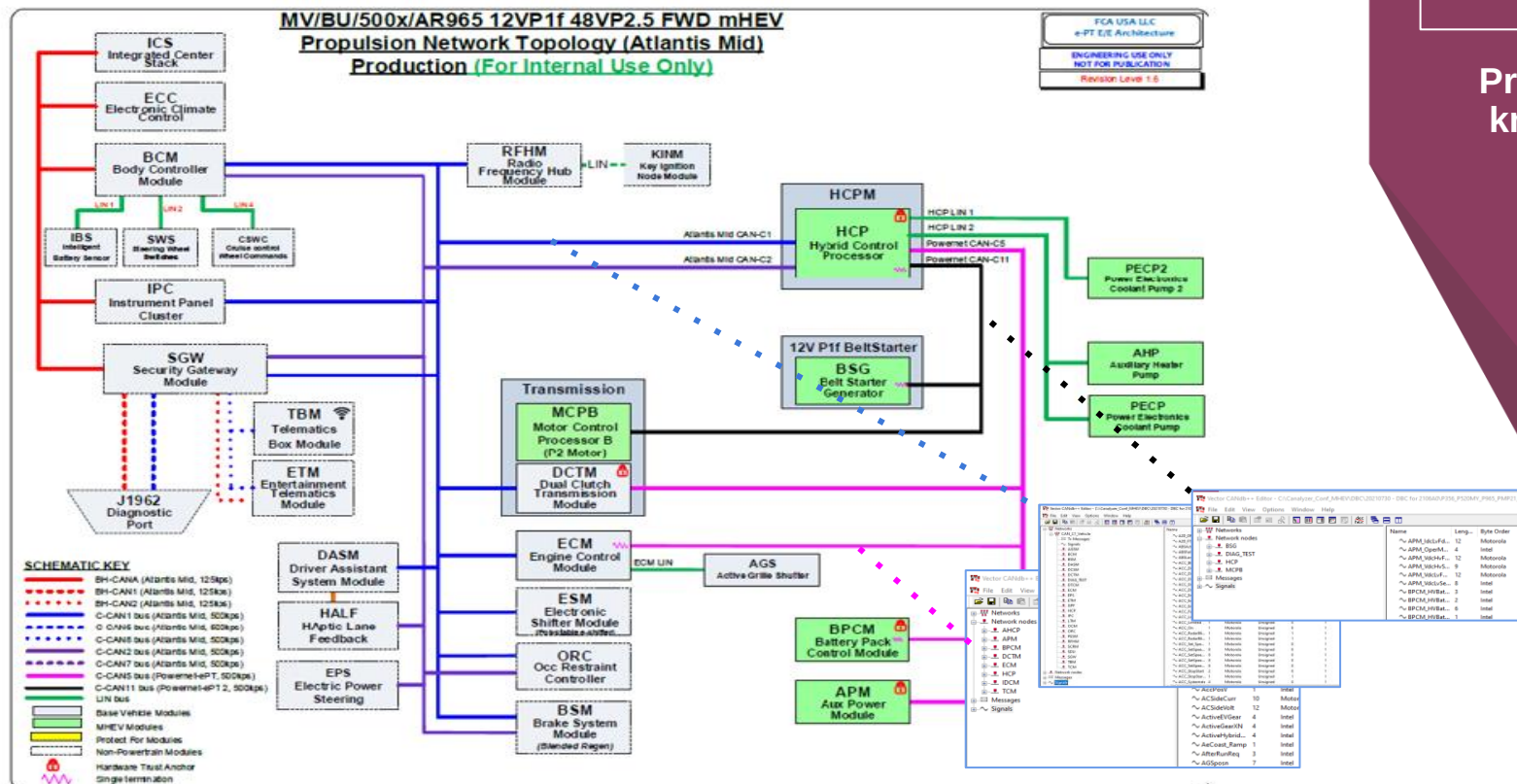
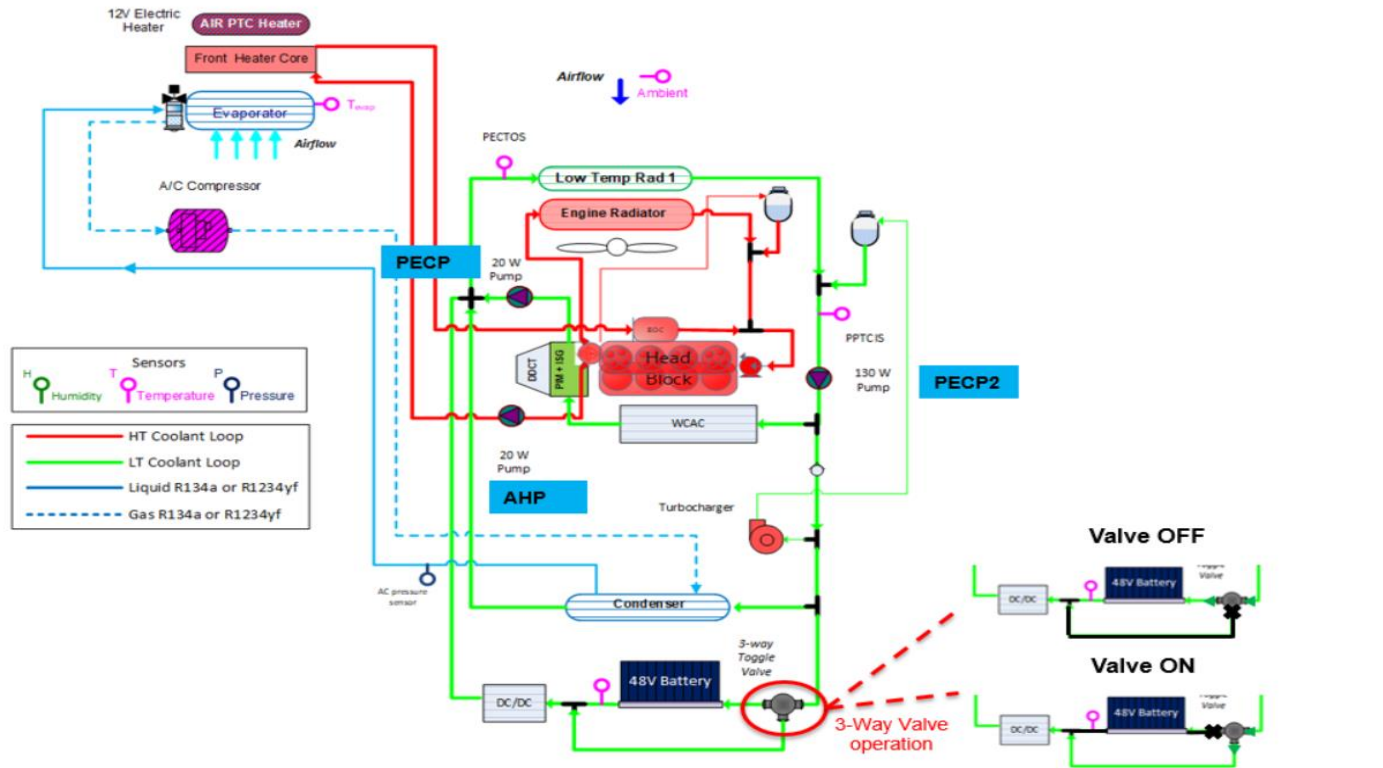


Figure 4 Vehicle Network Topology

Cooling System Configuration

Preliminary know-how 4/4





Case Study FLA - FTT#344 (MHEV vehicle)

FLA

Case Study
FTT#344Claim to Analyze: **Avaria Cambio**

A	B	C	D	E	F	G
Vehicle	Model	Fliotta	Descrizione anomalia	Data	Ora	Odo
PS127	334	RG	Mail Lorenzo Glorioso 03/02/2022 15:51 in Anticipo apertura sk i-Man a seguito accensione spia cambio e freno sulla PS127 per favore serve urgente diagnosi Mail Giuseppe Sangiovanni 03/02/2022 15:05 Ciao a tutti, la vettura in oggetto ha riscontrato la seguente anomalia: MT 14 24 - ODO 17458 km - Avaria cambio Driver interview: In avviamento, pochi metri dopo la partenza di inizio turno, si è accesa la spia di avaria cambio per elevata temperatura cambio.	03/02/2022	14:24	17458

Step 1 - Data availability check on controltec, download and if available compliance verification with VSR data and report from fleet engineer

APTIV QUALIFIER - SYSTEM VALIDATION SERVICE GMT 2022-04-04 13:48:21

HOME EVENTS SNAPSHOTS REPORTS VEHICLE DTC MANAGER ANALYTICS FILTERS

Package: All Packages Vehicle #: PS_0127 Fleet: All Fleets Build: Mile Pre YP Prod PS PS_HUC Test Status: Mileage Emissions Idle Overnight Completed Last Data Status: Current Orange Red Min Data Date: 2022-02-20 00:00 Max Data Date: 2022-02-03 12:18 Vehicle / VDR Search Apply Filters CLEAR FILTER

Vehicles VDRs Usage Properties Software

Vehicle Detail

VIN	Fleet	Veh No	Package	Location	Last Data	Build
ZFANF2VP0NP946461	RG_Pom	PS_0127	1.5 GSE4 MHEV / 33	NA_IT	2022-04-03	PS

Events DTC Modules RUMPR EV Charging EV Fuel Consumption

Event Detail: ZFANF2VP0NP946461

Date	Config	Event #	Event Name	Status	Fault Type	Odometer	Location	Comment
2022-02-03 22:23:40	4.44-19250	74056	Ch1-7_No_Busload_Alert_PHE	New	EV	17687	FCA EMEA Site	
2022-02-03 21:14:57	4.44-19250	74119	Manual Trigger	New	General	17628	FCA EMEA Site	
2022-02-03 18:44:48	4.44-19250	74056	Ch1-7_No_Busload_Alert_PHE	New	EV	17626	FCA EMEA Site	
2022-02-03 16:24:53	4.44-19250	74119	Manual Trigger	New	General	17532	FCA EMEA Site	
2022-02-03 16:21:36	4.44-19250	74056	Ch1-7_No_Busload_Alert_PHE	New	EV	17531	FCA EMEA Site	
2022-02-03 15:10:04	4.44-19250	74056	Ch1-7_No_Busload_Alert_PHE	New	EV	17492	FCA EMEA Site	
2022-02-03 14:25:25	4.44-19250	74119	Manual Trigger	New	General	17477	FCA EMEA Site	
2022-02-03 13:26:52	4.44-19250	74812	Open_Contactors_Vehicle_Sta	New	EV	17458	FCA EMEA Site	
2022-02-03 13:24:21	4.44-19250	74119	Manual Trigger	New	General	17456	FCA EMEA Site	
2022-02-03 13:24:15	4.44-19250	74612	Open_Contactors_Vehicle_Sta	New	EV	17455	FCA EMEA Site	
2022-02-03 13:18:21	4.44-19250	74119	Manual Trigger	New	General	17455	FCA EMEA Site	
2022-02-03 11:06:04	4.44-19250	74056	Ch1-7_No_Busload_Alert_PHE	New	EV	17445	FCA EMEA Site	
2022-02-03 10:49:48	4.44-19250	74119	Manual Trigger	New	General	17436	FCA EMEA Site	

EVENTS

Date	Event Name	Status	Fault Type	Odometer	Location
2022-02-03 13:24:21	Manual Trigger	New	General	17456	FCA EMEA Site
2022-02-03 13:24:15	Open_Contactors_Vehicle_Sta	New	EV	17455	FCA EMEA Site
2022-02-03 13:24:15	Manual Trigger	New	General	17455	FCA EMEA Site
2022-02-03 13:24:15	Ch1-7_No_Busload_Alert_PHE	New	EV	17455	FCA EMEA Site
2022-02-03 13:24:15	Manual Trigger	New	General	17455	FCA EMEA Site
2022-02-03 13:24:15	Ch1-7_No_Busload_Alert_PHE	New	EV	17455	FCA EMEA Site
2022-02-03 13:24:15	Manual Trigger	New	General	17455	FCA EMEA Site
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2022-02-03 13:24:15	Ch1-7_No_Busload_Alert_PHE	New	EV	17455	FCA EMEA Site

Step 2 - DTC Memory Check -> TCM first as reported by claim and then all other powertrain ECUs

Trigger Time	ECU	DTC	Status Byte	Desc	Status	Environmenta
2022-02-03 13:24:16	TCM	P0A3C-00	27	Drive Motor A Inverter Over Temperature	Lamp Off	+
2022-02-03 13:24:27	APM 11bit	P1ABC-00	0F	APM HV Battery System Voltage Low-	Lamp Off	+
2022-02-03 13:24:27	APM	P1ABC-00	0F	APM HV Battery System Voltage Low-	Lamp Off	+
2022-02-03 13:24:29	APM	P0607-00	0F	Control Module Performance	Lamp Off	+
2022-02-03 13:24:30	APM 11bit	P0607-00	0F	Control Module Performance	Lamp Off	+
2022-02-03 13:24:31	TCM	P0A3D-00	27	Drive Motor B Inverter Over Temperature	Lamp Off	+
2022-02-03 13:26:52	BPCM	P0C78-00	AE	Hybrid/EV Battery System Precharge Time Too Long	Lamp On	+
2022-02-03 13:26:53	APM 11bit	P1ABC-00	0E	APM HV Battery System Voltage Low-	Stored	
2022-02-03 13:26:53	BPCM	P0C78-00	2A	Hybrid/EV Battery System Precharge Time Too Long	Stored	
2022-02-03 13:26:53	APM	P1ABC-00	0E	APM HV Battery System Voltage Low-	Stored	
2022-02-03 13:26:53	APM 11bit	P0607-00	0E	Control Module Performance	Stored	
2022-02-03 13:26:53	APM	P0607-00	0E	Control Module Performance	Stored	
2022-02-03 13:27:01	TCM	P0A3C-00	26	Drive Motor A Inverter Over Temperature	Stored	
2022-02-03 13:27:14	TCM	P0806-00	2E	Clutch Position Sensor Circuit Range/Performance	New Stored	+

FLA

Case Study FTT#344

Step 3 - DTC Matrix check

CAN signals

- P2_Mode_Status =
Idle/Failure P0A3C on TCM
- P2_GMG_OVER_TEMPERATURE
error P0A3D on TCM

DTC Code	DTC Description	Warning Indicator FL/MIL (Note: MIL and ETC if ECM or	Maturation Failure Threshold
0x0A3C00	The inverter reports an error that affects the inverter or the e-machine. The Error excludes degradation according to overload / overtemperature (phase "black").	FL/MIL (3 trips)	inverter / e-machine error: MCPB_HYBRID_6.P2_Mode_Status is equal to Failure_ASC (0x8) OR MCPB_HYBRID_6.P2_Mode_Status is equal to Failure_Open (0x9) OR [idle state consideration is enabled via calibration AND MCPB_HYBRID_6.P2_Mode_Status is equal to Idle (0x0) AND ignition is on]
0x0A3D00	The inverter reports a degradation that affects the inverter or the e-machine (overtemperature).	FL/MIL (3 trips)	Inverter/ e-machine degradation: until HA351: HCP.MCPB_HYBRID_6.P2_INV_ThermalLoadrate_raw >= 120% OR HCP.MCPB_HYBRID_6.P2_EM_ThermalLoadrate_raw_if >= 120% from HA400: HCP.MCPB_HYBRID_2.P2_GMG_OVER_TEMPERATUR = 0x1 (Error)

Step 4 - CAN signals analysis

- Issue Visualization through signals
- Vehicle boundary conditions
- Possible root cause to address 2nd level

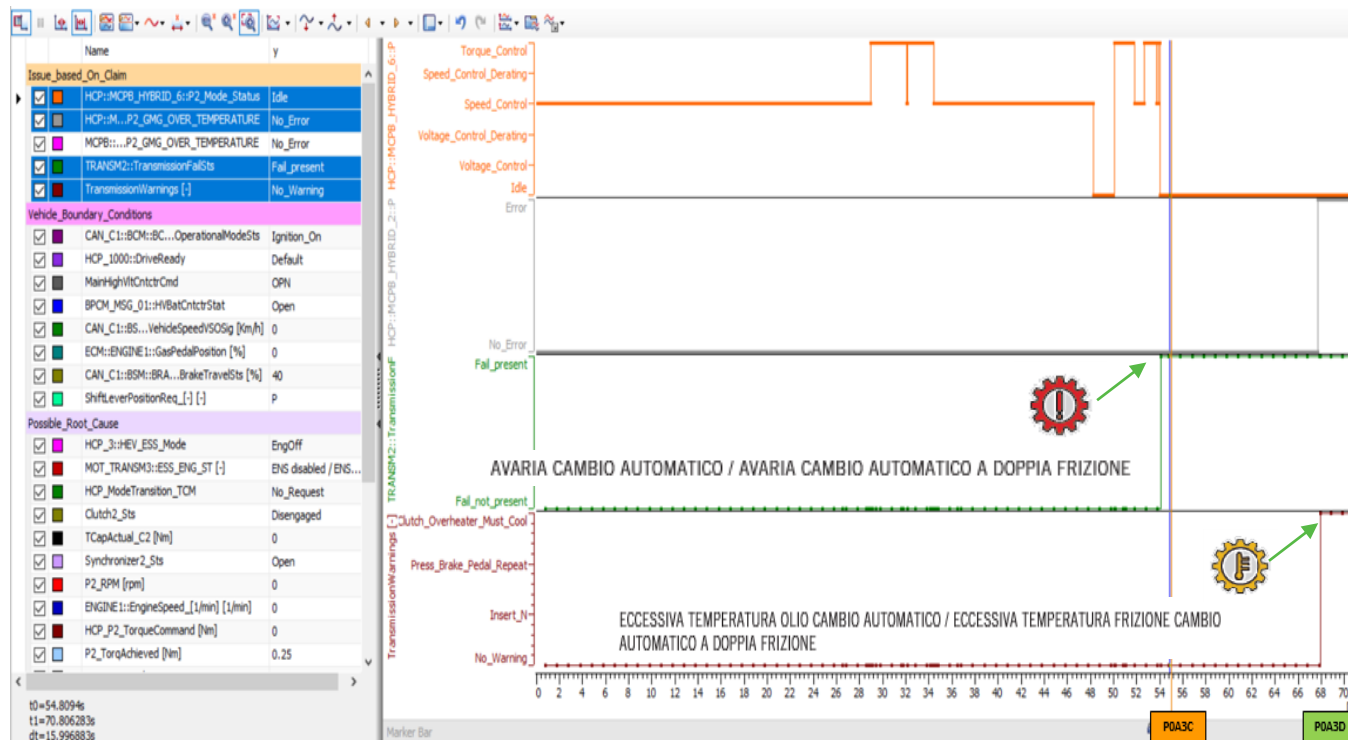
FLA

Case Study FTT#344

First goal of FLA is to identify the issue using CAN signals to have a graphical visual of driver's claim and which lamp or message has seen on the Instrument Panel Cluster (IPC).



Identify the issue is necessary to address the owner of 2nd level to analyze a specific point or range of time and not waste time analyzing data without any problem.

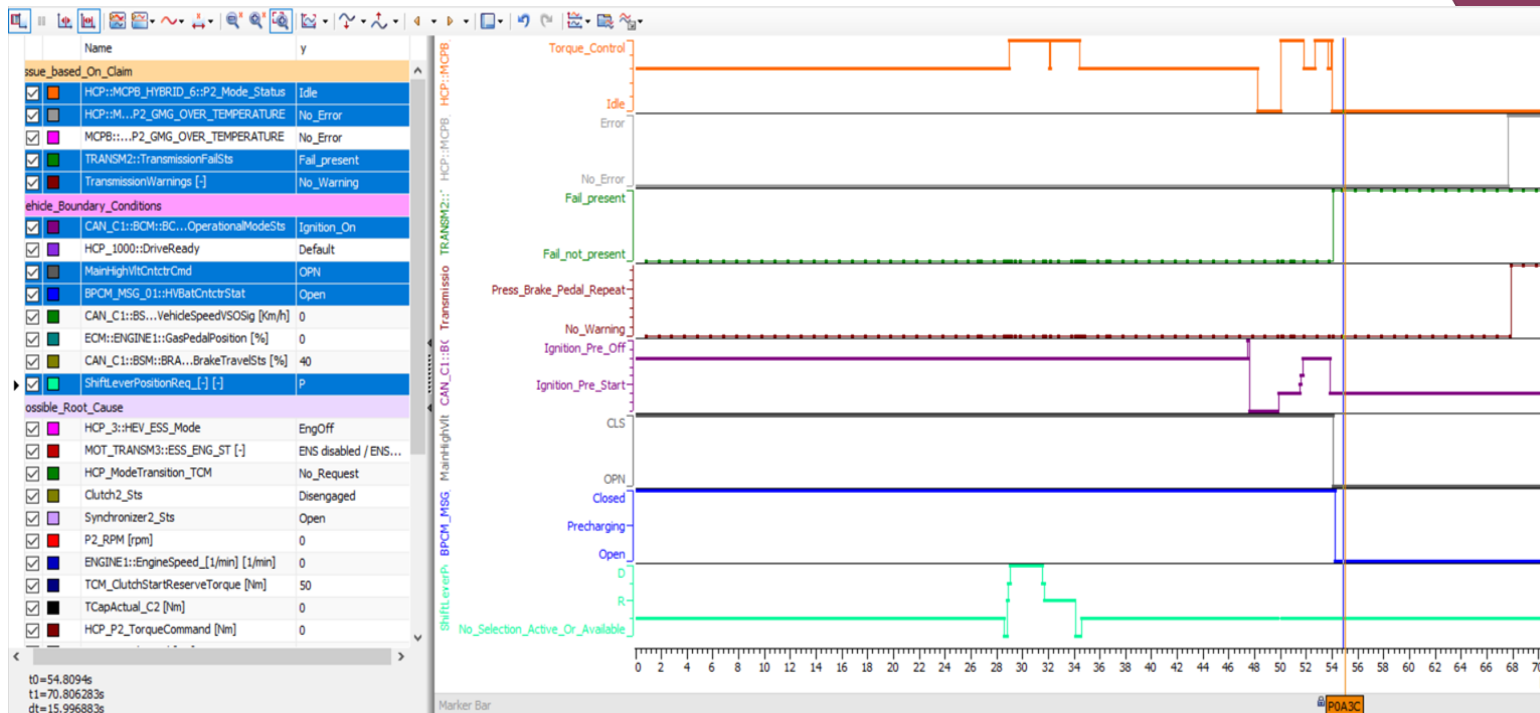


Step 4 - CAN signals analysis

- Issue Visualization through signals
- **Vehicle boundary conditions**
- Possible root cause to address 2nd level

FLA

Case Study FTT#344



Once found the issue it's useful to observe also the vehicle conditions when the anomaly occurred, to find if there was a specific maneuver executed by driver or not.

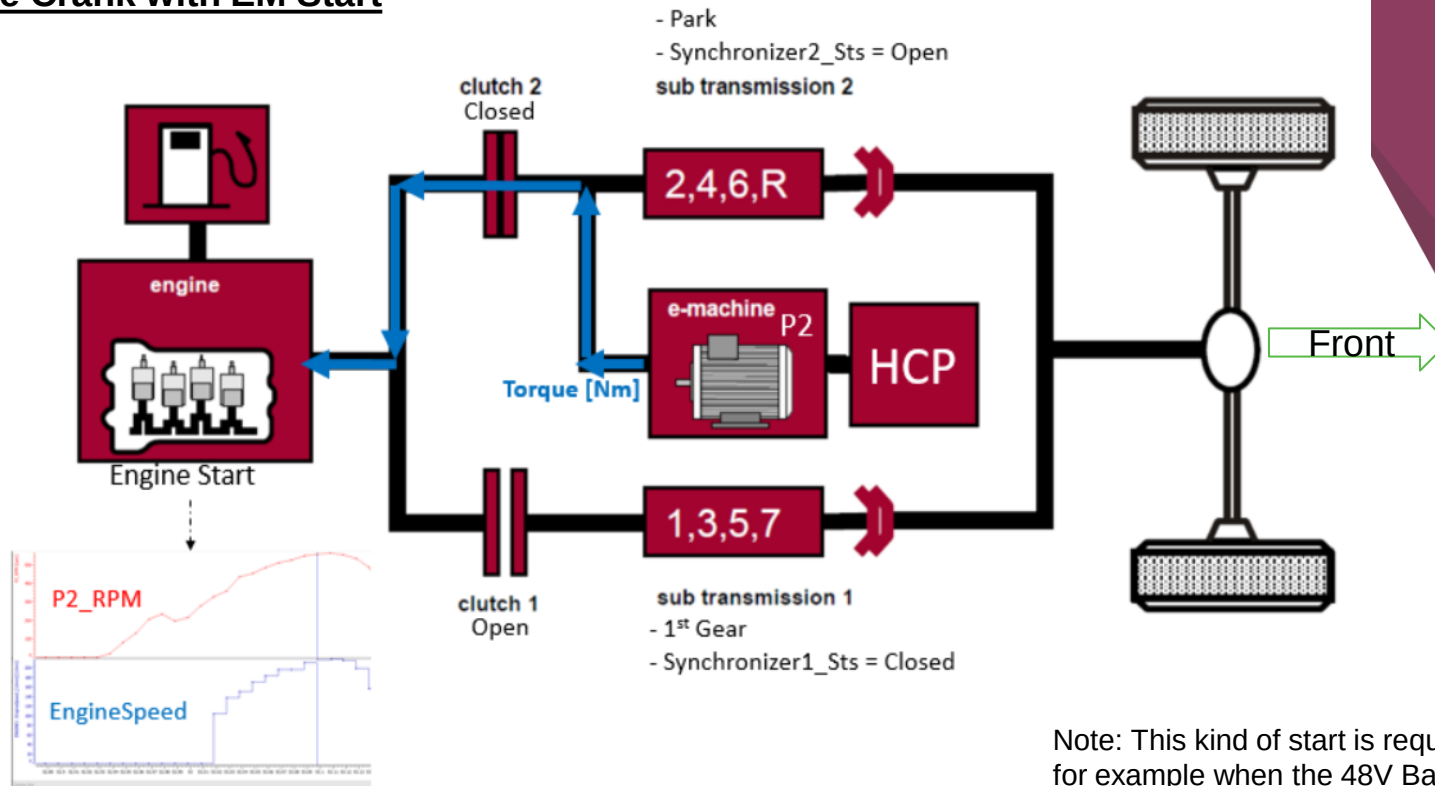
Boundary conditions are necessary especially when the root cause it's not found also by owner of 2nd level and start the activities of reappear test and monitoring of the issue.

Example:

- Drive Ready or KeyOn
- HEV or Hybrid
- Vehicle speed
- Engine Speed
- SOC 48V
- GasPedal
- Brake Pedal
- ShiftLeverPos
-

How it's possible to crank an ICE on a MHEV - 48V Vehicle?

Engine Crank with EM Start



Note: This kind of start is request for example when the 48V Battery SOC is less than 44.5 %.

FLA

Case Study
FTT#344

Step 4 - CAN signals analysis

- Issue Visualization through signals
- Vehicle boundary conditions
- Possible root cause to address 2nd level

FLA

Case Study FTT#344

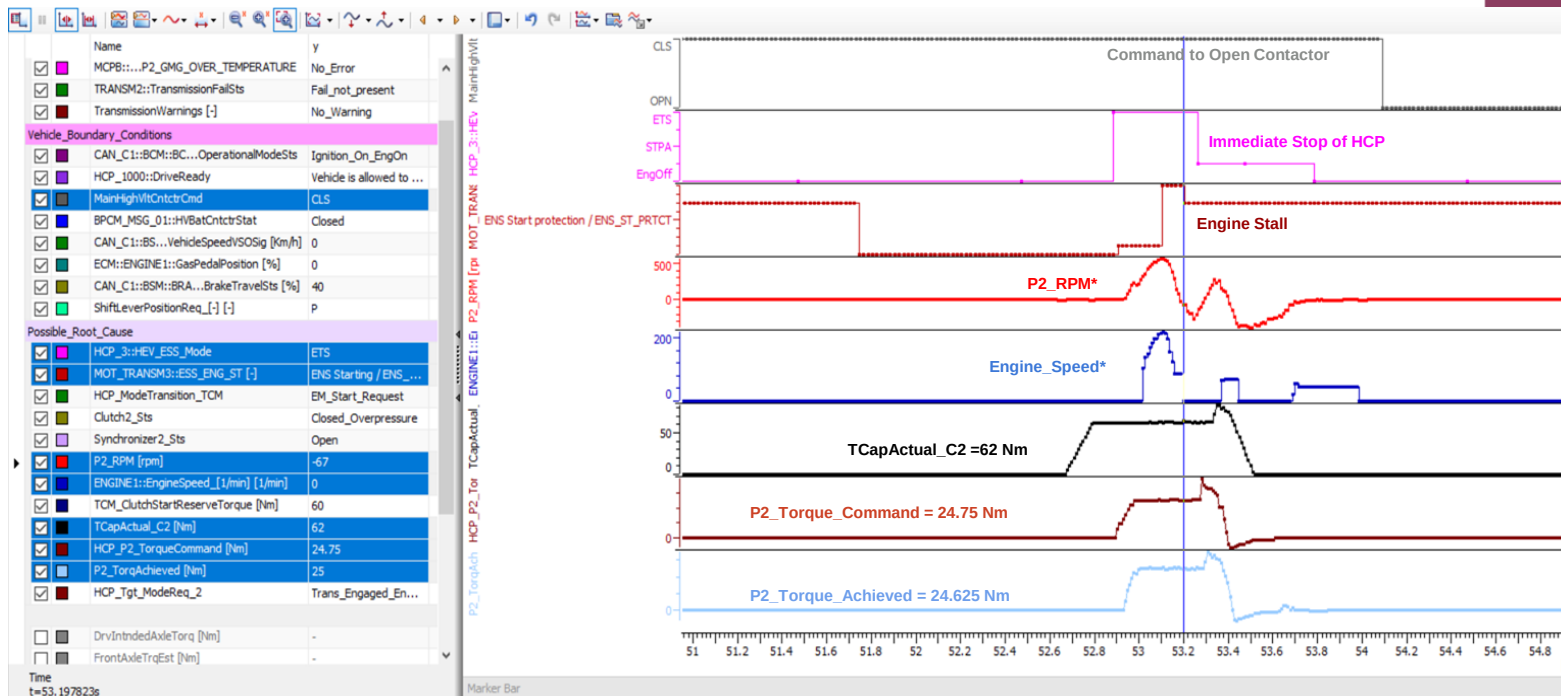
Issue visualization and boundary conditions confirm that DTCs in TCM and transmission lamp are consequence of an opening contactor after a crank of the vehicle.

"Why did the system opened the contactor?"

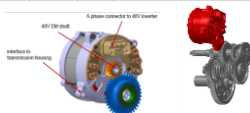
- The contactor opening was commanded by HCP after an engine stall during the crank (the ICE was not able to reach the combustion and the engine speed came back to zero)

"In which condition the engine declared stall?"

- The engine didn't start during an EM Low Power Start, despite 24 Nm of torque applied by P2 with clutch2 closed at 62 Nm



* intermediate gear ratio: 2,5 from P2 to SubTransmission2



The transmission lamp is due to the DTC P0A3C set by TCM after an opening contactors due to a Failed Engine Start with EM Low Power Start

Case Study FTT#344

- **ECM** for failed engine start with 24.75 Nm of torque applied by P2, more than 60 Nm at flywheel not enough to spin up the engine until the combustion
- **HCP** to confirm correct command to P2 during EM Low Power Start and reaction to engine stall



Step 6 - FTT MHEV update - Program Validation Manager Task

- 02/09: IFLD shared
- 03/04: Issue related to a low EM torque delivered for ICE start in warm condition (over 50°C coolant temperature). Issue will be fixed on TCM with an higher saturation offset which will be set as a minimum saturation during EM start and with higher torque delivered on 200 and 400 engine rpm. Possible introduction on a dedicated PCR6B4 (plan under definition)

FLA

Case Study
FTT#344

1	FTT#	CAN LOG	Open ▾	Severity	Entry Date	W solution date	in Validation date	Close date	Application	Vehicle	VIN	Common/ Unique	PCR
345	344	https://drive.google.com/drive/folders/1L8YjRRCF58rxzmmHPm9U64D8FgMV5Dkn	With Solution ▾	50 ▾	9-Feb-2022				334	PS117			PCR5E

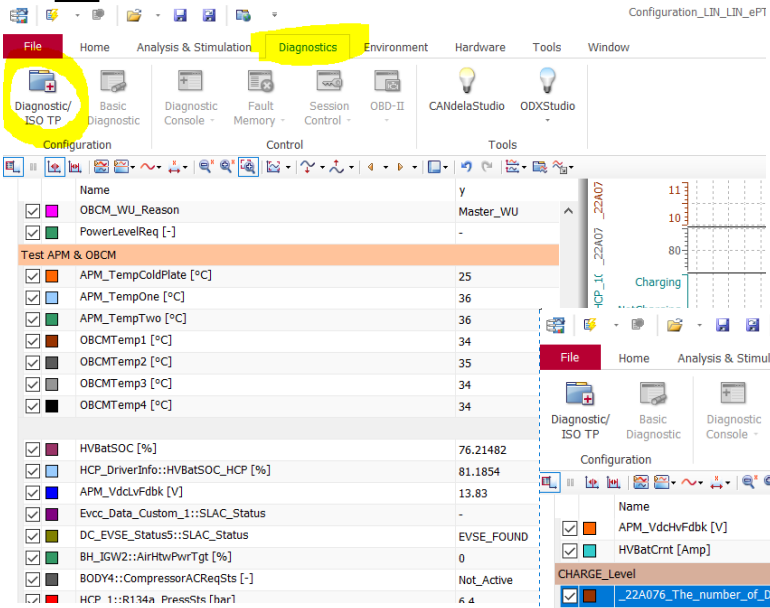
Note from chain mail and meeting after FTT#344 opening:

- HCP confirmed correct reaction to engine stall -> immediate stop and opening contactor -> P0A3C dtc and TransmissionLamp
- HCP confirmed torque command to P2 -> it's calculated from the TCapActual_C2 (torque of closing Clutch2 signal of TCM)
- ECM confirm that torque was not enough to spin up more the engine and start the combustion (60 Nm at flywheel)
- Improvement: **TCM SW calibration change**: higher torque applied by TCM on Clutch2 during EM Start to avoid engine stall (P2 torque will be calculated in HCP from signal TCapActual_C2).



PDX use in validation & troubleshooting

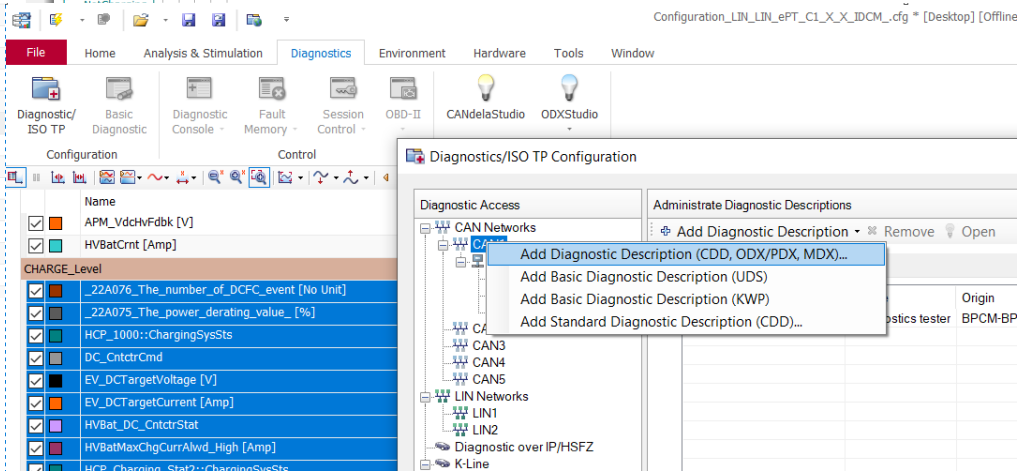
1



FLA

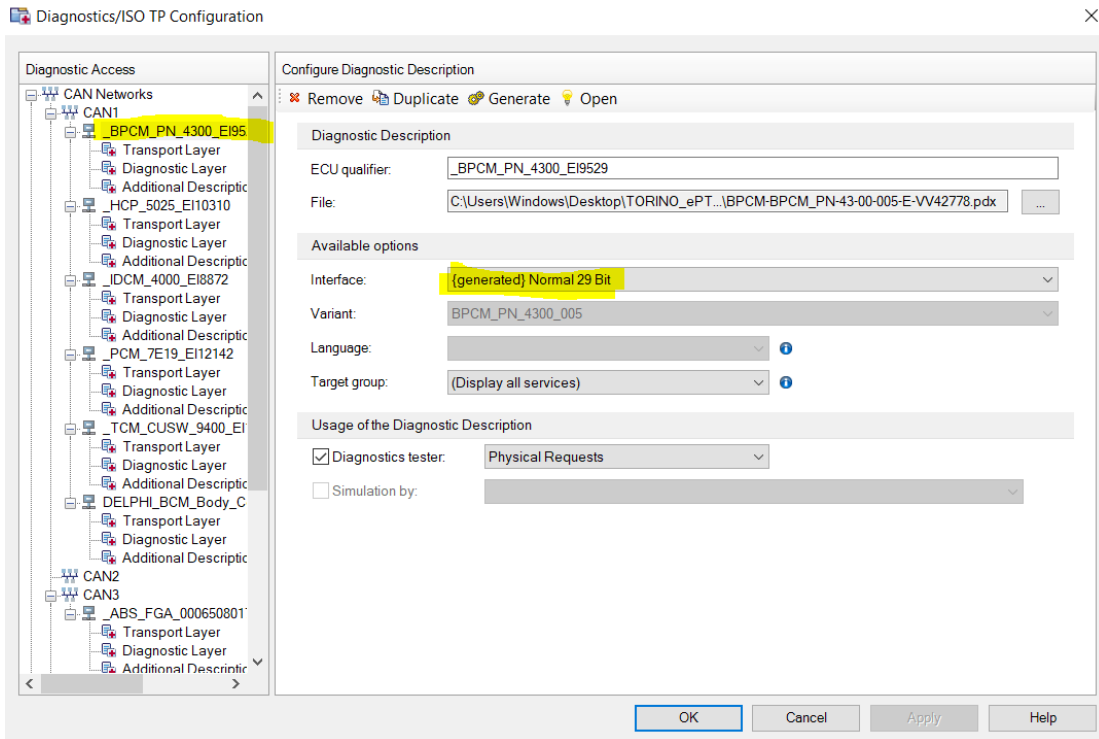
Diagnostic
Response
Decoding

2: Select the can
that contains C1 and
Diagnostic messages



3: Select the can that contains C1 and Diagnostic messages

Select Interface (Ex: Normal 29 Bit for CAN C1 – Atlantis)



FLA

Diagnostic
Response
Decoding

4: Check IDs for the BPCM diagnostic in «trasport layer tab»

FLA

Diagnostic
Response
Decoding

Diagnostics/ISO TP Configuration

Diagnostics/ISO TP Configuration

Diagnostic Access

CAN Networks

CAN1

_BPCM_7900_EI12222_1

Transport Layer

Diagnostics Layer

Additional Descriptions

CAN2

CAN3

CAN4

CAN5

LIN Networks

LIN1

LIN2

Diagnostic over IP/HSFZ

K-Line

COM

Additional Diagnostic Settings

ISO TP Observer

ISO TP Advanced

DolP/HSFZ Main Settings

Configure Transport Layer (CAN1)

Parameters from diagnostic description for transport and diagnostic layer

☒ Override manually ⓘ

Reset to defaults

Addressing

	Mode	CAN ID (hex)	29 bit identifier	ECU	Tester	TX prio
Request to ECU	NormalFixed	0x18DA44F1	☒	0x44	0xF1	0x6
Response from ECU	NormalFixed	0x18DAF144	☒	0x44	0xF1	0x6
Functional to ECU	NormalFixed	0x18DB02F1	☒	0x2	0xF1	0x6
UUDT from ECU	UUDT	0x780	☒	0x0	0x0	0x0

Additional ISO TP protocol parameters

STmin: 20 ms

Block size: 0

FC delay: 10 ms

Max. length: 4095

OK

Cancel

Apply

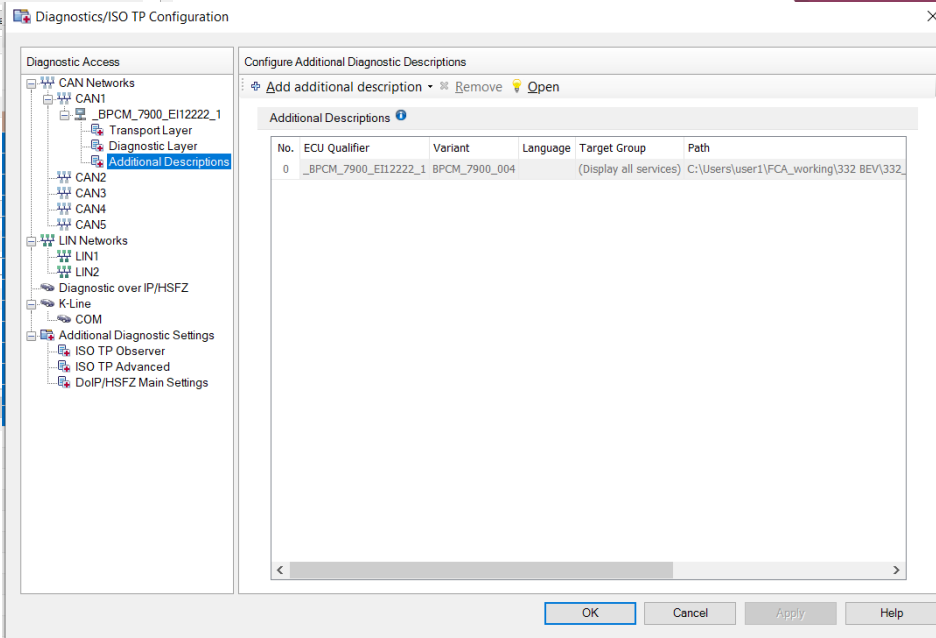
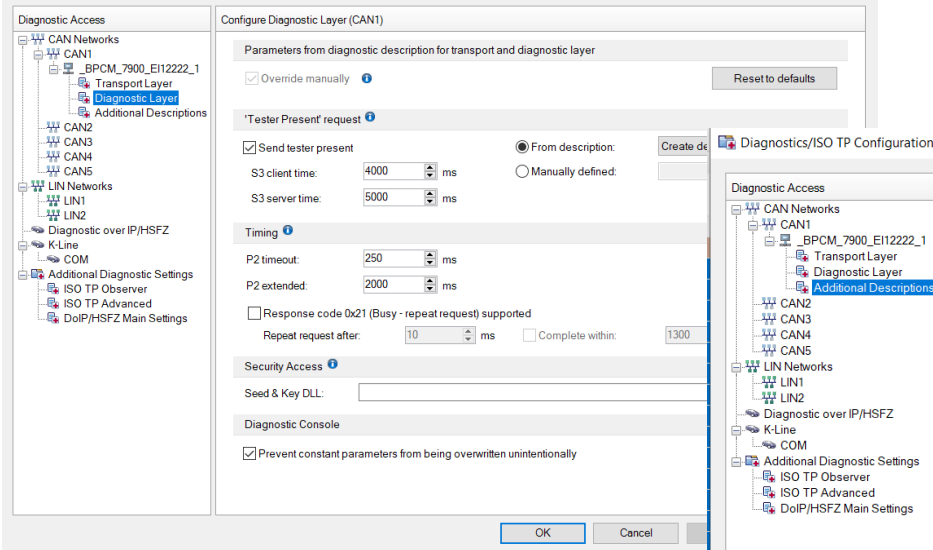
Help

5: Check and then ok

FLA

Diagnostic
Response
Decoding

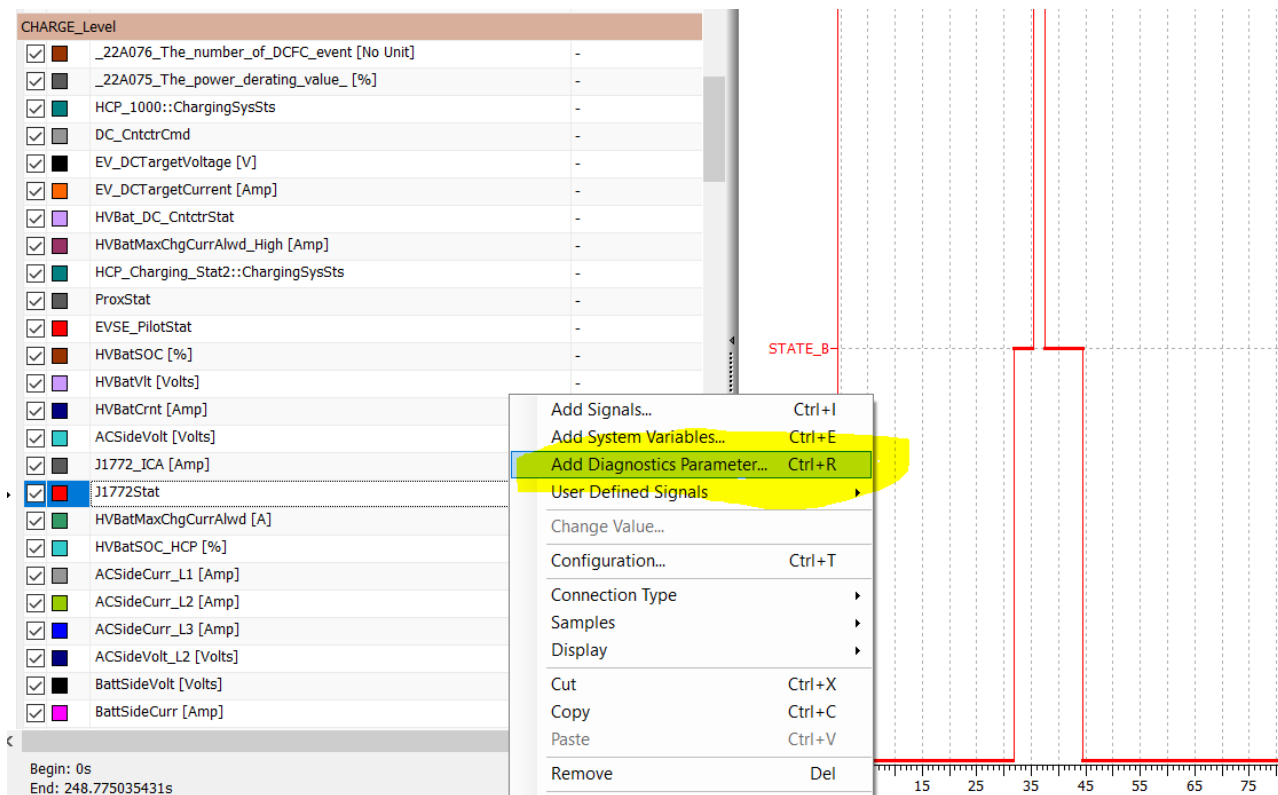
Diagnostics/ISO TP Configuration



6: You can add the RDI and others in graph with Ctrl + R

FLA

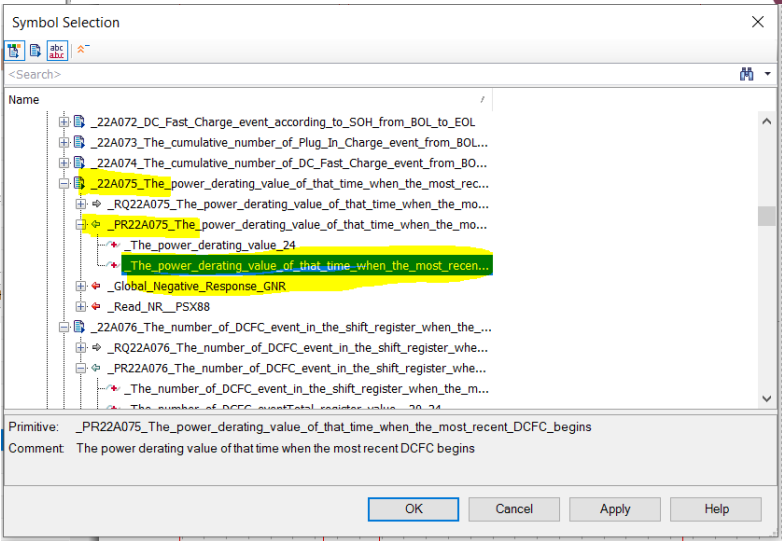
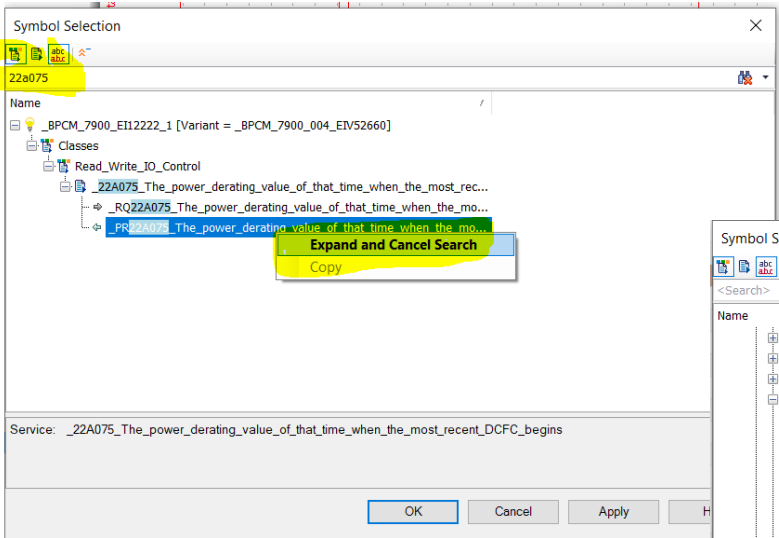
Diagnostic
Response
Decoding



7: Pick what needed

FLA

Diagnostic
Response
Decoding

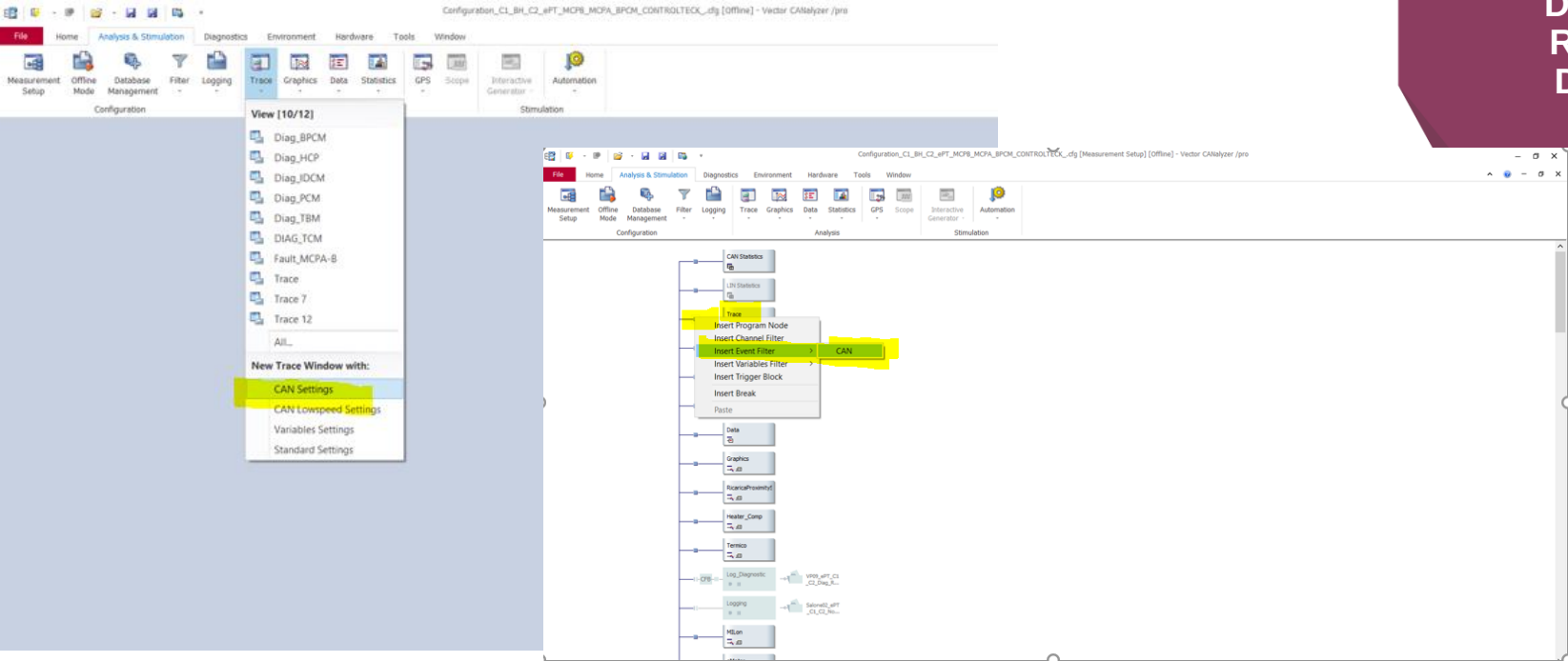


Canalyzer – Diagnostic Trace

1: Create a new trace and filter only diagnostic message

FLA

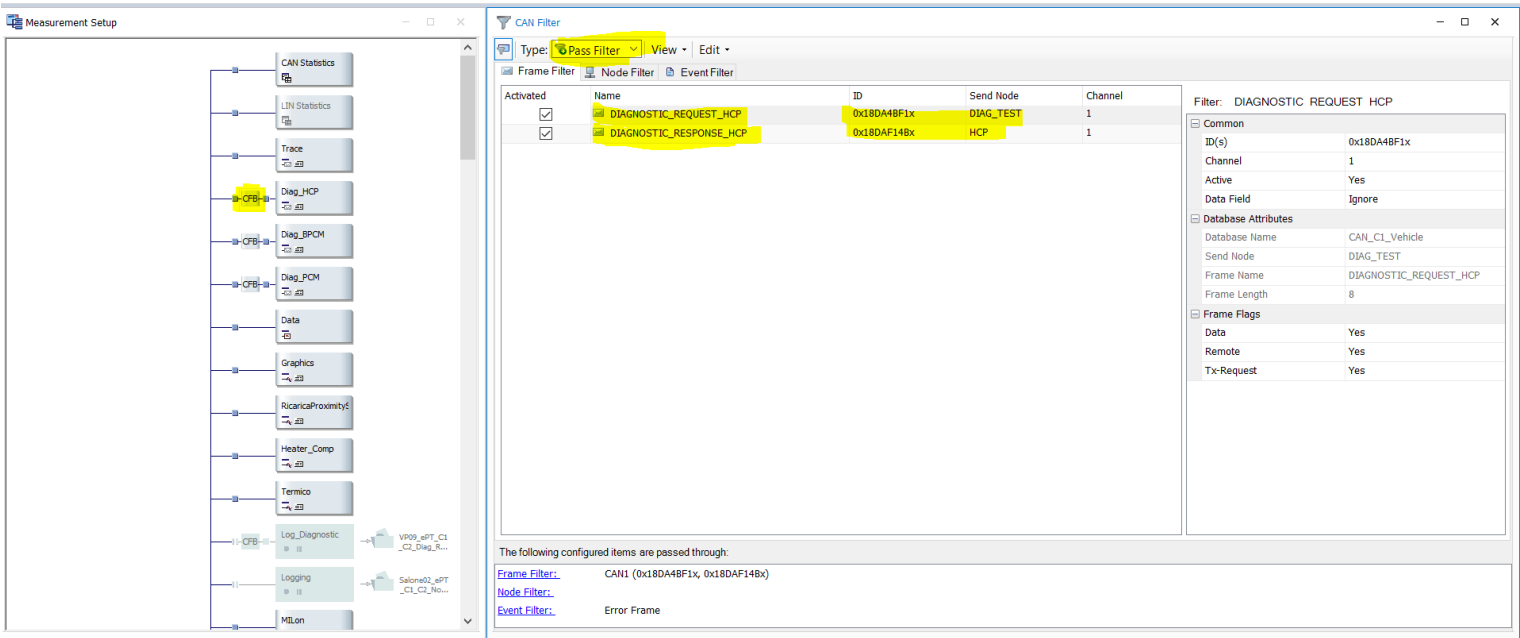
Diagnostic
Response
Decoding



2: Select in Pass Filter Diagnostic Message you need

FLA

Diagnostic
Response
Decoding



3: Open the trace created and filter for ID message used in the filter

FLA

Diagnostic
Response
Decoding

The screenshot shows the Canalyzer Measurement Setup window. On the left, a tree view lists various diagnostic components, with 'Diag_HCP' highlighted. The main window displays the 'Diag_HCP' trace, showing a table with columns: Time, Chn, ID, Sender N..., Name, Event Type, Dir, DLC, Data length, and Data. A 'Custom Column Filter' dialog box is open, showing the filter configuration for the ID column. The filter is set to 'equals' and '18DA4BF1x'. The 'Show rows' section shows 'ID' with 'equals' and '18DAF14Bx'.

4: All diagnostic message Rx-Tx also in hex value

FLA

Diagnostic
Response
Decoding

Configuration_C1_BH_C2_ePT_MCPB_MCPA_BPCM_CONTROLTECK_cfg * [Diag_HCP] [Offline] - Vector CANalyzer /pro

File Home Analysis & Stimulation Diagnostics Environment Hardware Tools Window

Measurement Setup Offline Mode Database Management Filter Logging Trace Graphics Data Statistics GPS Scope Interactive Generator Automation

Configuration Analysis Simulation

Time	Chn	ID	Sender N...	Name	Event Type	Dir	DLC	Data length	Data
0.237574	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 19 02 00 [00 00 00 00]
0.237574	CAN 1	18DA4BF1x		Report DTC By Status Mask::req	req		3		19 02 00
0.244188	CAN 1	18DAF148x	HCP	DIAGNOSTIC_RESPONSE_HCP	SF	Rx	7	8	[07] 59 02 FF 0C F3 00 68
0.244188	CAN 1	18DAF148x		Report DTC By Status Mask::pos	pos		7		59 02 FF 0C F3 00 68
0.568703	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 22 30 33 [00 00 00 00]
0.568703	CAN 1	18DA4BF1x		Motor A Inverter Current Offset::req	req		3		22 30 33
0.585112	CAN 1	18DAF148x	HCP	DIAGNOSTIC_RESPONSE_HCP	SF	Rx	6	7	[06] 62 30 33 80 80 80
0.585112	CAN 1	18DAF148x		Motor A Inverter Current Offset::pos	pos		6		62 30 33 80 80 80
0.649427	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 22 30 34 [00 00 00 00]
0.649427	CAN 1	18DA4BF1x		Motor B Inverter Current Offset::req	req		3		22 30 34
0.674624	CAN 1	18DAF148x	HCP	DIAGNOSTIC_RESPONSE_HCP	SF	Rx	6	7	[06] 62 30 34 7F 7F 7F
0.674624	CAN 1	18DAF148x		Motor B Inverter Current Offset::pos	pos		6		62 30 34 7F 7F 7F
0.714334	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 22 30 40 [00 00 00 00]
0.714334	CAN 1	18DA4BF1x		Motor A Resolver Angle Offset::req	req		3		22 30 40
0.736400	CAN 1	18DAF148x	HCP	DIAGNOSTIC_RESPONSE_HCP	SF	Rx	4	5	[04] 62 30 40 3E
0.736400	CAN 1	18DAF148x		Motor A Resolver Angle Offset::pos	pos		4		62 30 40 3E
0.774351	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 22 30 41 [00 00 00 00]
0.774351	CAN 1	18DA4BF1x		Motor B Resolver Angle Offset::req	req		3		22 30 41
0.795457	CAN 1	18DAF148x	HCP	DIAGNOSTIC_RESPONSE_HCP	SF	Rx	4	5	[04] 62 30 41 50
0.795457	CAN 1	18DAF148x		Motor B Resolver Angle Offset::pos	pos		4		62 30 41 50
0.1246193	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 19 02 00 [00 00 00 00]
0.1246193	CAN 1	18DA4BF1x		Report DTC By Status Mask::req	req		3		19 02 00
0.1250637	CAN 1	18DAF148x	HCP	DIAGNOSTIC_RESPONSE_HCP	SF	Rx	7	8	[07] 59 02 FF 0C F3 00 68
0.1250637	CAN 1	18DAF148x		Report DTC By Status Mask::pos	pos		7		59 02 FF 0C F3 00 68
0.1568730	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 22 30 33 [00 00 00 00]
0.1568730	CAN 1	18DA4BF1x		Motor A Inverter Current Offset::req	req		3		22 30 33
0.1585616	CAN 1	18DAF148x	HCP	DIAGNOSTIC_RESPONSE_HCP	SF	Rx	6	7	[06] 62 30 33 80 80 80
0.1585616	CAN 1	18DAF148x		Motor A Inverter Current Offset::pos	pos		6		62 30 33 80 80 80
0.1650820	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 22 30 34 [00 00 00 00]
0.1650820	CAN 1	18DA4BF1x		Motor B Inverter Current Offset::req	req		3		22 30 34
0.1671396	CAN 1	18DAF148x	HCP	DIAGNOSTIC_RESPONSE_HCP	SF	Rx	6	7	[06] 62 30 34 7F 7F 7F
0.1671396	CAN 1	18DAF148x		Motor B Inverter Current Offset::pos	pos		6		62 30 34 7F 7F 7F
0.1709557	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 22 30 40 [00 00 00 00]
0.1709557	CAN 1	18DA4BF1x		Motor A Resolver Angle Offset::req	req		3		22 30 40
0.1725311	CAN 1	18DAF148x	HCP	DIAGNOSTIC_RESPONSE_HCP	SF	Rx	4	5	[04] 62 30 40 3E
0.1725311	CAN 1	18DAF148x		Motor A Resolver Angle Offset::pos	pos		4		62 30 40 3E
0.1776316	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 22 30 41 [00 00 00 00]
0.1776316	CAN 1	18DA4BF1x		Motor B Resolver Angle Offset::req	req		3		22 30 41
0.1797750	CAN 1	18DAF148x	HCP	DIAGNOSTIC_RESPONSE_HCP	SF	Rx	5	8	[04] 62 30 41 50
0.1797750	CAN 1	18DAF148x		Motor B Resolver Angle Offset::pos	pos		4		62 30 41 50
0.2251390	CAN 1	18DA4BF1x	DIAG_TEST	DIAGNOSTIC_REQUEST_HCP	SF	Tx	3	8	[03] 19 02 00 [00 00 00 00]
0.2251390	CAN 1	18DA4BF1x		Report DTC By Status Mask::req	req		3		19 02 00

Select Toggle Display Mode
(all messages in timeline)

5: Filter Data length (Empty) to have only Tx-Rx Translated Msg (green)

FLA

Diagnostic
Response
Decoding

Configuration_C1_BH_C2_ePT_MCPB_MCPA_BPCM_CONTRUITECK_cdg * [Diag_HCP] [Offline] - Vector CANalyzer / pro

File Home Analysis & Stimulation Diagnostics Environment Hardware Tools Window

Measurement Setup Offline Mode Database Management Filter Logging Trace Graphics Data Statistics GPS Scope Interactive Generator Automation

Configuration Analysis Stimulation

<Search>

Time	Chn	ID	Sender N...	Name	Event Type	Dir	DLC	Data length	Data
0.000000								(All)	
0.237574	CAN 1	180A4BF1x		Report DTC By Status Mask::req	req		3	(Custom...)	0 68
0.244188	CAN 1	180AF14Bx		Report DTC By Status Mask::pos	pos		7	(Empty)	
0.568703	CAN 1	180A4BF1x		Motor A Inverter Current Offset::req	req		3		
0.585112	CAN 1	180AF14Bx		Motor A Inverter Current Offset::pos	pos		6		
0.649427	CAN 1	180A4BF1x		Motor B Inverter Current Offset::req	req		3		
0.674624	CAN 1	180AF14Bx		Motor B Inverter Current Offset::pos	pos		6		
0.714334	CAN 1	180A4BF1x		Motor A Resolver Angle Offset::req	req		3		
0.739400	CAN 1	180AF14Bx		Motor A Resolver Angle Offset::pos	pos		4		
0.774351	CAN 1	180A4BF1x		Motor B Resolver Angle Offset::req	req		3		
0.795457	CAN 1	180AF14Bx		Motor B Resolver Angle Offset::pos	pos		4		
1.246193	CAN 1	180A4BF1x		Report DTC By Status Mask::req	req		3		19 02 00
1.250637	CAN 1	180AF14Bx		Report DTC By Status Mask::pos	pos		7		58 02 FF 0C F3 00 68
1.568730	CAN 1	180A4BF1x		Motor A Inverter Current Offset::req	req		3		22 30 33
1.585616	CAN 1	180AF14Bx		Motor A Inverter Current Offset::pos	pos		6		62 30 33 80 80 80
1.650820	CAN 1	180A4BF1x		Motor B Inverter Current Offset::req	req		3		22 30 34
1.671396	CAN 1	180AF14Bx		Motor B Inverter Current Offset::pos	pos		6		62 30 34 7F 7F 7F
1.709557	CAN 1	180A4BF1x		Motor A Resolver Angle Offset::req	req		3		22 30 40
1.725311	CAN 1	180AF14Bx		Motor A Resolver Angle Offset::pos	pos		4		62 30 40 3E
1.776316	CAN 1	180A4BF1x		Motor B Resolver Angle Offset::req	req		3		22 30 41
1.797750	CAN 1	180AF14Bx		Motor B Resolver Angle Offset::pos	pos		4		62 30 41 50
2.251390	CAN 1	180A4BF1x		Report DTC By Status Mask::req	req		3		19 02 00
2.256710	CAN 1	180AF14Bx		Report DTC By Status Mask::pos	pos		7		58 02 FF 0C F3 00 68
2.567901	CAN 1	180A4BF1x		Motor A Inverter Current Offset::req	req		3		22 30 33
2.587404	CAN 1	180AF14Bx		Motor A Inverter Current Offset::pos	pos		6		62 30 33 81 80 80
2.651730	CAN 1	180A4BF1x		Motor B Inverter Current Offset::req	req		3		22 30 34
2.671357	CAN 1	180AF14Bx		Motor B Inverter Current Offset::pos	pos		6		62 30 34 7F 7F 7F
2.717539	CAN 1	180A4BF1x		Motor A Resolver Angle Offset::req	req		3		22 30 40
2.738001	CAN 1	180AF14Bx		Motor A Resolver Angle Offset::pos	pos		4		62 30 40 3E
2.778610	CAN 1	180A4BF1x		Motor B Resolver Angle Offset::req	req		3		22 30 41
2.800104	CAN 1	180AF14Bx		Motor B Resolver Angle Offset::pos	pos		4		62 30 41 50
3.256369	CAN 1	180A4BF1x		Report DTC By Status Mask::req	req		3		19 02 00
3.260677	CAN 1	180AF14Bx		Report DTC By Status Mask::pos	pos		7		58 02 FF 0C F3 00 68
3.569773	CAN 1	180A4BF1x		Motor A Inverter Current Offset::req	req		3		22 30 33
3.591175	CAN 1	180AF14Bx		Motor A Inverter Current Offset::pos	pos		6		62 30 33 80 80 80
3.653745	CAN 1	180A4BF1x		Motor B Inverter Current Offset::req	req		3		22 30 34
3.670708	CAN 1	180AF14Bx		Motor B Inverter Current Offset::pos	pos		6		62 30 34 7F 7F 7F
3.713369	CAN 1	180A4BF1x		Motor A Resolver Angle Offset::req	req		3		22 30 40
3.728337	CAN 1	180AF14Bx		Motor A Resolver Angle Offset::pos	pos		4		62 30 40 3E
3.781121	CAN 1	180A4BF1x		Motor B Resolver Angle Offset::req	req		3		22 30 41
3.801416	CAN 1	180AF14Bx		Motor B Resolver Angle Offset::pos	pos		4		62 30 41 50
4.253450	CAN 1	180A4BF1x		Report DTC By Status Mask::req	req		3		19 02 00

Canalyzer – Diagnostic Trace & Marker in Graphics

FLA

Diagnostic
Response
Decoding

The screenshot displays the Vector CANalyzer interface. The top menu bar includes File, Home, Analysis & Stimulation, Diagnostics, Environment, Hardware, Tools, and Window. The left sidebar shows the Configuration pane with tabs for Measurement Setup, Offline Mode, Database Management, Filter, Logging, Trace, Graphics, Data, Statistics, GPS, Scope, Interactive Generator, and Automation. The main workspace is divided into two panes. The top pane shows a list of diagnostic messages with columns for Time, Chn, ID, Sender N., Name, Event Type, Dir, DLC, Data length, and Data. The bottom pane shows a tree view of the diagnostic messages, with a selected message expanded to show its details. A dialog box titled 'Edit Marker' is open, showing the marker name 'A. t = 56.40608s' and various attributes like Name, Background color, Time, and Comment.

Time	Chn	ID	Sender N.	Name	Event Type	Dir	DLC	Data length	Data
55.770193	CAN 1	18DA4BF1x		Motor B Inverter Current Offset::req	req		3	22 30 34	
55.790205	CAN 1	18DAF148x		Motor B Inverter Current Offset::pos	pos		6	62 30 34 7F 80 7F	
55.839666	CAN 1	18DA4BF1x		Motor A Resolver Angle Offset::req	req		3	22 30 40	
55.869478	CAN 1	18DAF148x		Motor A Resolver Angle Offset::pos	pos		4	62 30 40 3E	
55.918588	CAN 1	18DA4BF1x		Motor B Resolver Angle Offset::req	req		3	22 30 41	
55.940034	CAN 1	18DAF148x		Motor B Resolver Angle Offset::pos	pos		4	62 30 41 3E	
56.403566	CAN 1	18DA4BF1x		Report DTC By Status Mask::req	req		3	19 02 00	
56.406080	CAN 1	18DAF148x		Report DTC By Status Mask::pos	pos		7	59 02 FF 0C F3 00 68	

Variant: "HCP_5025_010_ENV46625"

- Read DTC Information Positive Response
- Report DTC By Status Mask
- DTC Status Availability Mask
- DTC and status of DTC
 - EndOfServiceIndicator0
 - DTC
 - Status of DTC
 - Test Failed
 - Test Failed This Operation Cycle
 - Pending DTC
 - Confirmed DTC
 - Test Not Completed Since Last Clear
 - Test Failed Since Last Clear
 - Test Not Completed This Operation Cycle
 - Warning Indicator Requested

0000: 59 02 FF 0C F3 00 68 Y.....h

Motor A Inverter Current Offset::req

Motor A Inverter Current Offset::pos

Motor B Inverter Current Offset::req

Motor B Inverter Current Offset::pos

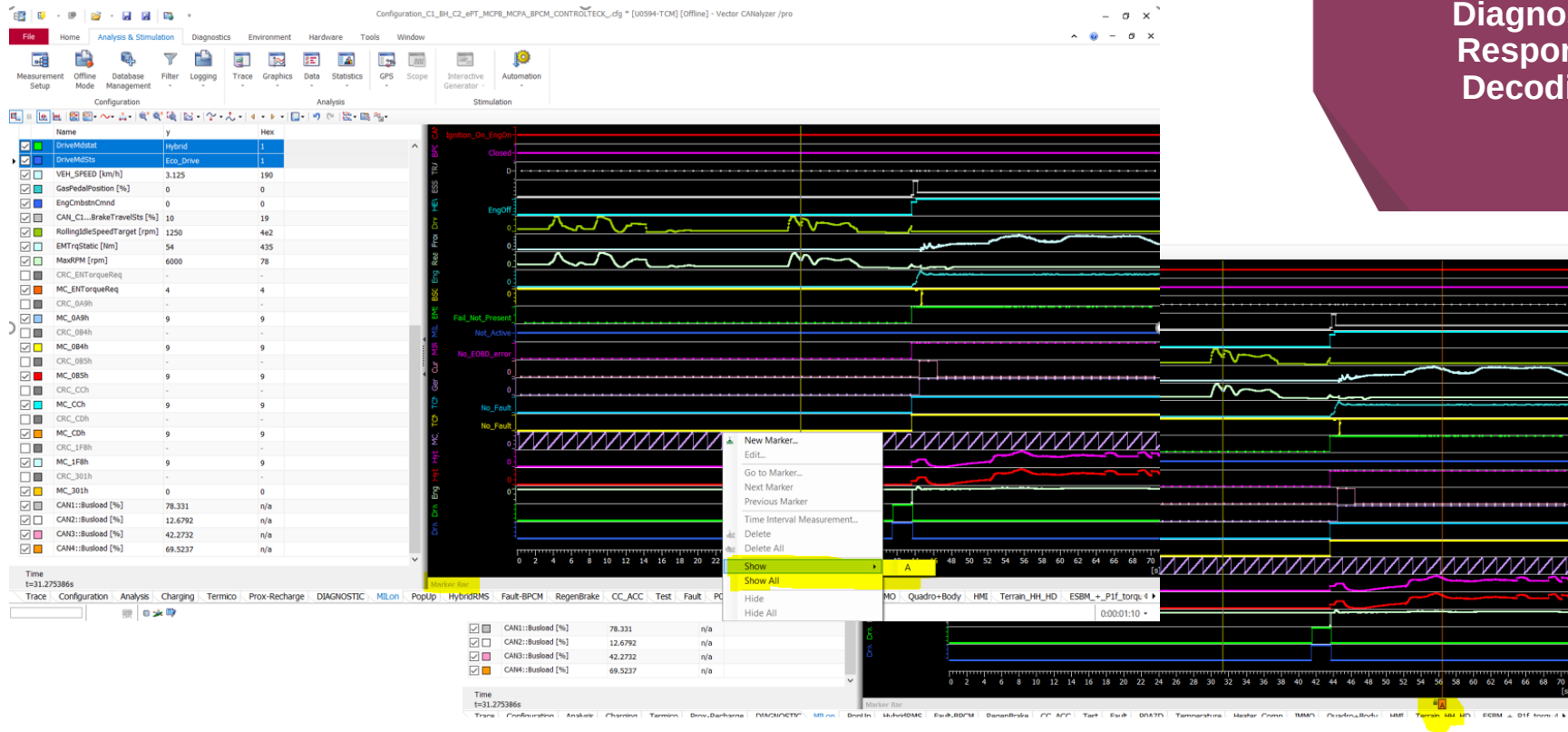
Motor A Resolver Angle Offset::req

Canalyzer – Diagnostic Trace & Marker in Graphics

2: Right mouse button on marker bar and show the marker you want

FLA

Diagnostic
Response
Decoding



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